Sameer Albostamy

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Education

McMaster University Sept 2023 – April 2027
Bachelor of Arts in Computer Science and Economics *Hamilton, ON*

- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Software Development, Computer Architecture, Software Design, Databases, Linear Algebra, Intermediate Economics, Statistics

Work Experience

Software Developer Intern | Python, Java, SQL, PostgreSQL, Dext September 2025 – December 2025
Support Ontario Youth *Toronto, ON*

- Automated quarterly funder reporting by building Python and SQL pipelines across **1,400+** apprentice and employer records, cutting report prep time by **35%**.
- Wrote Python validation scripts to clean and deduplicate JotForm intake data, improving data accuracy by **30%** ahead of government submissions.
- Automated Dext expense exports reconciliation against accounting records in Java, reducing manual finance review time by **45%**.
- Documented **23** standardized finance policies from existing accounting workflows, enabling consistent review and faster staff onboarding.

Software Developer Intern | Python, JavaScript, Flask, React, MySQL May 2025 – August 2025
Media Phoenix *Toronto, ON*

- Built and maintained client websites using HTML, CSS, Flask, and JavaScript, improving cross-browser compatibility and mobile responsiveness.
- Wrote Python scripts to automate repetitive web maintenance tasks, reducing manual effort across **15+** client projects.
- Set up a MySQL database to store client data and wrote scripts to pull inputs from web analytics tools.

Technical Projects

Vanguard-AI: Clinical Predictive Risk Engine | Repo | Python, Scikit-Learn, NumPy, Pandas, Matplotlib January 2026

- Trained a Random Forest Classifier on **2,000+** synthetic patient records to predict Acute Kidney Injury risk, achieving **92%** accuracy on held-out test data.
- Built a pharmacokinetic ODE solver using Euler integration to simulate drug accumulation across multi-dose cycles, enabling precise therapeutic window tracking.
- Reduced simulated AKI flagging latency by **80%** by vectorizing patient batch processing with NumPy over a naive loop implementation.
- Architected end-to-end risk pipeline from dose input to real-time Matplotlib violation alerts, cutting manual review workload by **60%** in testing scenarios.

McMaster Enrolment Automation Engine | Repo | Python, Selenium, Chrome WebDriver July 2025

- Engineered a Selenium-based automation engine serving **50+** concurrent users with a **100%** course registration success rate at open enrollment.
- Optimized async element selection and dynamic wait-states, reducing per-user registration time from **8** minutes to under **45** seconds.
- Integrated structured exception handling and logging, achieving zero-fault execution across **200+** live registration runs.

Technical Skills

Languages: Python, Java, C/C++, SQL, HTML/CSS, R

Technologies: React, Flask, PostgreSQL, MySQL, SQLite, Docker, Selenium, Pandas, NumPy, Scikit-Learn, Matplotlib, Excel, Power BI, Linux, Git, GitHub

Concepts: Object-Oriented Design, Functional Programming, REST API Design and Integration, Automation and Scripting, Web Development (Frontend & Backend), Test Automation, Data Modeling, Machine Learning, Software Architecture, Version Control, Performance Optimization

Interests: Soccer, UFC, Weightlifting, Cars, Chess, Aviation